CURRICULUM VITAE

KAILANI P. ALARCON

University of Nevada, Reno · Graduate Study, Electrical Engineering

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EDUCATION

University of Nevada, Reno | Graduate Study in Electrical Engineering Fall 2026 – Present

Graduate coursework in progress: Independent Study in Control Systems (EE 793E), Principles of Computer Operating Systems (CS 646), and Game Theory for Cybersecurity (CS 703).

University of Nevada, Reno | B.S. in Electrical Engineering August 2026

Emphasis: Robotics, Autonomous/Aerial Vehicles, and Embedded Systems; Minor in Mathematics; Honors Baccalaureate (conferred August 2026).

Selected coursework: Control Systems I and II, Introduction to Robotics, Signals and Systems, Digital Signal Processing, Linear Algebra, Probability and Statistics, Embedded Systems Design, and Microprocessor Applications.

RESEARCH FOCUS

Research interests: Reliable robotic manipulation under degraded vision and variable contact, with a focus on sensor fusion, state estimation, and adaptive impedance control. NASA lunar-dust arm feasibility work motivated questions about when a robot should adjust its estimates or contact behavior to complete tasks safely.

RESEARCH EXPERIENCE

NASA Marshall Space Flight Center | Office of STEM Engagement (OSTEM) Intern Aug 2026 – Present

Human Factors Engineering and Mechanical Design | Technical mentor: Elizabeth (Liz) Bowman-Langley | Huntsville, AL

- Independently evaluated a robotic arm for proposed lunar-dust mitigation tests in high vacuum, analyzing geometry, joint travel, reachable workspace, loading, and actuator compatibility.
- Calculated an estimated 3.1 N·m shoulder self-load moment for a horizontal arm configuration; identified excluded rocker and actuator masses requiring further center-of-gravity analysis before actuator selection.
- Developed 34 hierarchical system requirements from six operating scenarios addressing manipulation, vacuum compatibility, workspace constraints, emergency shutdown, and verification.
- Developed preliminary control software and atmospheric bench-test procedures for the existing Pololu Micro Maestro 6-channel controller; documented verification needs for proposed vacuum-compatible actuators.

University of Nevada, Reno | Three-Link Human-Leg Dynamics Jan – May 2025

Honors Signature Experience | Faculty mentor: Dr. Sesh Commuri | Student-initiated, faculty-mentored research

- Developed a MATLAB Level-2 S-function for a planar three-link leg model in Simulink; computed foot position, Cartesian velocity, and approximate joint torques from hip, knee, and ankle states.
- Implemented forward kinematics and Jacobian velocity mapping; formulated an inertia matrix and used Coriolis/centrifugal and gravitational terms in simplified joint-torque calculations.

University of Nevada, Reno | Virtual Vector Network Analyzer for Stepped-Frequency Radar Jan – May 2025

EE 330 Honors Contract / Honors Signature Experience | Faculty mentor: Dr. Jihwan Yoon | Student-initiated project

- Developed a MATLAB-based two-port virtual vector network analyzer (VNA) workflow for stepped-frequency continuous-wave (SFCW) radar, with 236 measurement points from 100 MHz to 6 GHz.
- Implemented 12-term short-open-load-through (SOLT) calibration, corrected-versus-uncorrected S-parameter visualization, and automated Touchstone (.s2p) exports.

TECHNICAL PUBLICATIONS AND PRESENTATIONS

Industry White Papers

1. Alarcon, K. P. (2026, February 5). *Thermal Validation of Commercial LED Luminaires.* LUX Dynamics.
2. Alarcon, K. P. (2026, March 10). *Understanding Optical Lens Performance in LED Luminaires.* LUX Dynamics.

Posters

1. *IVerify: Noninvasive IV Occlusion Detection System.* Poster presented at Innovation Day, May 2026. Senior capstone design and preliminary validation.

2. *Designing LEGO: Environmental Sustainability and Engineering Design Review.* Poster presented at the ENGR 301 Poster Session, University of Nevada, Reno, May 2025.

Industry white papers and student posters are identified by type; they are not represented as peer-reviewed journal or conference articles.

SELECTED ENGINEERING PROJECTS

IVerify: Noninvasive IV Occlusion Detector | Senior Capstone Spring 2026

Project and systems integration lead for a four-person team. Coordinated system architecture, requirements, integration, and documentation across the clamp, force sensor, analog front end, microcontroller, and indicator subsystems. Integrated a proof-of-concept bench prototype with repeatable force-response sensing, approximately 5 N clamp preload, and two-LED status indication; organized requirements-to-test traceability.

UR3e Robot Kinematics | Robotics Team Project Spring 2026

Modeled robot kinematics using Denavit-Hartenberg transforms, MATLAB Jacobians, and singularity analysis; evaluated a configuration with Jacobian rank five.

HueMotion: Camera-Based Gesture Control | LUX Dynamics 2026

Developed a real-time MediaPipe hand-tracking interface using gesture stabilization and two-hand recognition to trigger RGBW lighting scenes; extended control from wired DMX hardware to wireless Casambi API scene activation.

Fingerprint-Based Authentication | Embedded Systems Team Project Fall 2025

Designed and integrated an STM32 fingerprint authentication prototype in a four-person team, using UART sensor communication, enrollment and identification of 24 templates, packet handling, and status indicators.

PROFESSIONAL EXPERIENCE

LUX Dynamics | Product Development Intern Sep 2023 – Aug 2026

Reno, NV

- Conducted electrical, thermal, and photometric evaluations for more than 12 LED products, informing product-validation and manufacturing-release decisions.
- Independently developed a team-adopted Python/Arduino thermal-validation workflow with eight Pico TC-08 thermocouple channels, programmable setpoints, heater control, stability checks, and automated reporting.
- Developed a tablet-based RGBW lighting quality-control interface using Arduino Mega and DMX512; authored two publicly available LUX Dynamics technical white papers.

Sonic Drive-In | Associate Manager Aug 2020 – Aug 2023

Reno-Sparks, NV

Managed daily operations, training, scheduling, and compliance while pursuing engineering studies.

TEACHING, MENTORING AND OUTREACH

Engineering Peer Tutor | University of Nevada, Reno Aug 2024 – May 2026

Engineering Study Hub / Engineering Tutoring Center

Tutored approximately 12 to 15 engineering students weekly in circuits, signals, controls, embedded C, STM32 microcontrollers, MATLAB, and Simulink.

Teacher Assistant Intern | University of Nevada, Reno Nov 2024 – May 2025

Academic Support Program

Assisted with grading, academic support, course logistics, and communication among students, mentors, and program leadership.

E-FIT Mentor | University of Nevada, Reno May – Aug 2025

Academic Support Program

Guided incoming engineering students during university transition activities.

Honors College Ambassador | University of Nevada, Reno May – Aug 2025

Honors College Outreach

Assisted with campus tours, recruitment events, and student orientation.

AWARDS, HONORS AND SCHOLARSHIPS

- August 2026 — Honors Baccalaureate, University of Nevada, Reno
- May 2026 — Outstanding Service Award, College of Engineering

- May 2026 — Outstanding Service Award, Department of Electrical Engineering
- May 2026 — Innovation Day Finalist, IVerify Senior Capstone Team
- January 2026 — TC Energy Scholarship
- May 2025 — Honors College Global Travel Award
- August 2022 — Presidential Scholarship, University of Nevada, Reno

INTERNATIONAL ACADEMIC EXPERIENCE

Università degli Studi della Tuscia | Honors Abroad with USAC, Italy May – Jul 2024

Coursework: Italian language, photography, and art crime (USAC Honors Abroad).

TECHNICAL SKILLS

Robotics and controls: MATLAB, Simulink, robot kinematics and Jacobians, rigid-body dynamics, PID and lead-lag controllers, actuator loading; hands-on ROS 2 and MuJoCo.

Programming and embedded systems: Python, C/C++, VHDL, STM32, Arduino, Linux, Git; UART, SPI, I²C, PWM, ADC, GPIO, and interrupt-driven design.

Engineering tools: Creo, STM32CubeIDE, Tkinter, Flask, PyQt6, Excel-based analysis and reporting; hands-on Thermal Desktop; introductory AutoCAD and SolidWorks.

Measurement and verification: Oscilloscopes, thermocouples/Pico TC-08, thermal and photometric testing, radar S-parameter calibration, requirements development, and test planning.

LANGUAGES

English: fluent | Spanish: intermediate (speaking, reading, writing) | Italian: intermediate speaking; beginner reading and writing | Samoan: beginner | Russian: beginner

PROFESSIONAL AFFILIATIONS

Member: Institute of Electrical and Electronics Engineers (IEEE); Society of Women Engineers (SWE); American Society of Mechanical Engineers (ASME).